FR201x-U Datasheet

High-performance and low-power Bluetooth MCU

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Description

FR201x-U is a family of SOC (system on chip) for rapid development of Bluetooth Low Energy related products. It contains Bluetooth V5.3 (LE Mode) fully compliant system with Freqchip designed firmware and software stack. Users can develop various applications based on the built-in high performance 32-bit CPU.

The Bluetooth Smart firmware includes the L2CAP service layer protocols, Security Manager (SM), Attribute Protocol (ATT), the Generic Attribute Profile (GATT) and the Generic Access Profile (GAP). Furthermore, application profiles such as Proximity, Health Thermometer, Heart Rate, Blood Pressure, Glucose, Human Interface Device (HID) and SDK (include drivers, OS API, etc.) are supported.

With Freqchip's innovational technology, FR800x-U integrates, PMU, QSPI flash ROM with XIP mode, I2C, UART, GPIO, SARADC, PWM all in a single chip, which provides customer with:

1. competitive power consumption
2. stable connection
3. low-cost BOM

Features

CPU and On-chip Memory

- Embedded 32-bit CPU
  - Clock frequency: 12MHz–96MHz
- Built-in 1 independent DMA controller
- 256KB ~ 512KB Flash
- 32KB SRAM

Bluetooth

- Compliant with Bluetooth Specification V5.3 LE
- RX Sensitivity: -97dBm (LE 1Mbps)
- RX Sensitivity: -94dBm (LE 2Mbps)
- RX Sensitivity: -99dBm (LE S2)
- RX Sensitivity: -100dBm (LE S8)
- TX Power -10dBm ~10dBm
- Supports dynamic TX power control
- Supports data rates of 2M/1M/500K/125K

Interface

- Digital Interface
  - Up to GPIO*36
  - Timer *2
  - SPIM *2 (x1/x2/x4)
  - SPIS
  - UART *3
  - I2C *2
  - PWM *8
  - PDM
  - USB2.0 FS
  - 8080/6800 parallel interface
  - RGB565 Display Interface
  - True Random Number Generator (TRNG)
  - AES-128 Hardware Encryption and Decryption Engine
- Analog Interface
  - 9-channel 10bit SAR ADC
    - Maximum sampling data rate 500K
    - Supports periodic sampling

Power Management

- Integrates Buck and LDO regulators for powering each on-chip module
- Integrates power switches for different power domains
- Low-power Parameters (Buck Mode, 3.3 V, 25 °C)
  - Overall Bluetooth TX/RX Current
    - Typical RX Current: 6.01 mA
    - Typical TX Current @ 0 dBm: 7.6 mA
  - 3.3V Supply, Sleep Mode with RTC

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Enabled, Wake-up via Any GPIO, 40KB SRAM Retention: 5.4 μA
- Power-off Mode: 2.3 μA @ SRAM Non-retention
- Shutdown Mode: 80 nA
- Average Power Consumption at 1s Broadcast Interval (BLE): 21.8 μA (Tx Power 0 dBm)
- Average Power Consumption at 1s Connection Interval (BLE): 16.9 μA (Tx Power 0 dBm)

Operating Conditions:

- Operating ambient temperature: -40°C ~ +105°C

Applications

- Smart Keyboard and Mouse
- Smart Wearable
- Smart Home
- Smart Mobility
- Industrial Internet of Things (IIoT)

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Ordering Information

Part Number	Frequency	FLASH	GPIO	Ambient temperature	operating supply voltage	Package	Size	MSL
FR2011B-U	96MHz	256KB	12	-40°C ~ +105°C	2.0V~3.6V	QFN20	3.0×3.0×0.75, 0.4 pitch	MSL3
FR2012A-U	96MHz	512KB	24	-40°C ~ +105°C	2.0V~3.6V	QFN32	4.0*4.0*0.75, 0.4 pitch	MSL3
FR2018A-U	96MHz	512KB	36	-40°C ~ +105°C	2.0V~3.6V	QFN48	6.0×6.0×0.75, 0.4pitch	MSL3

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1. Hardware Details

1.1 Block Diagram

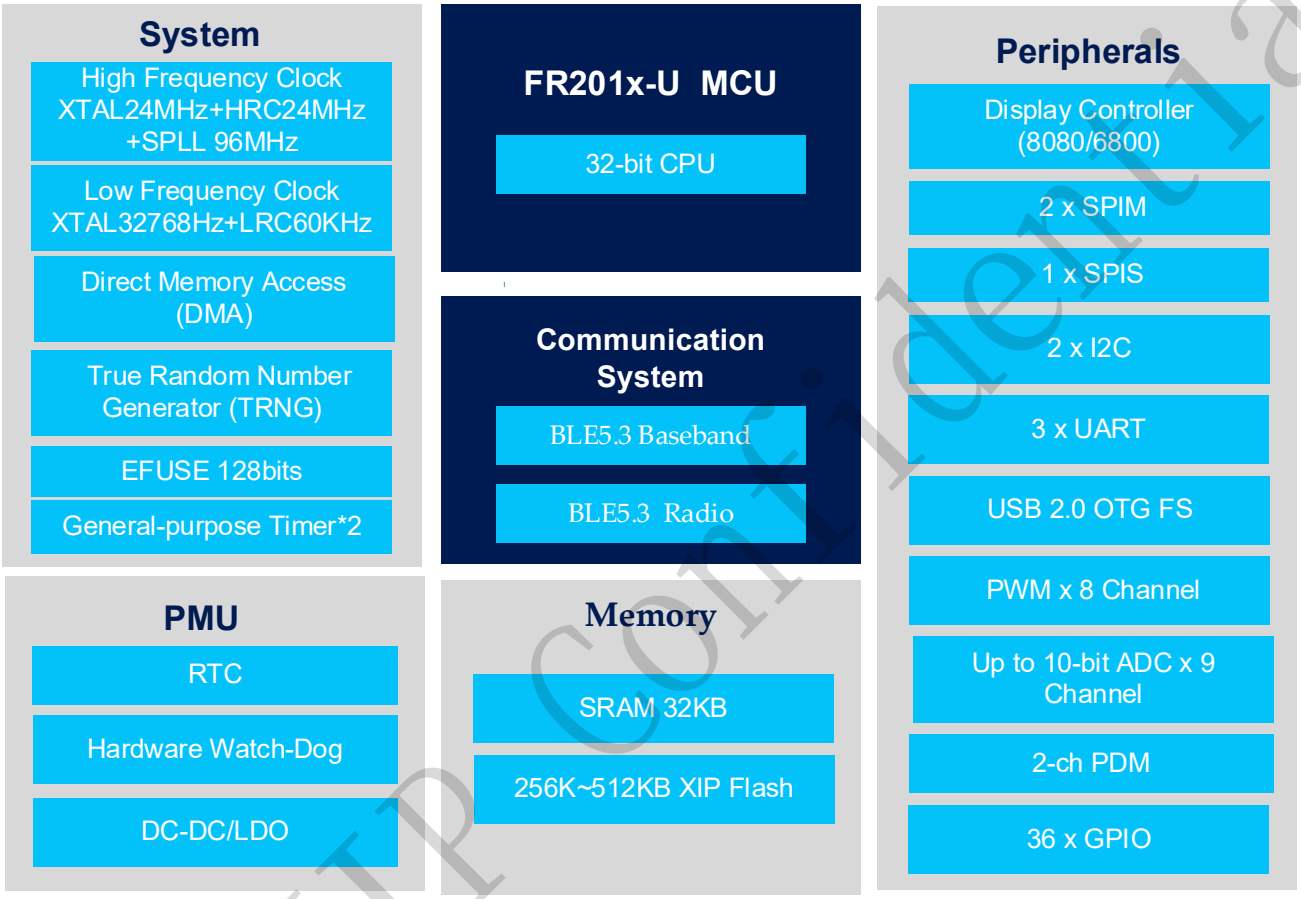


Figure 1-1 Block Diagram

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1.2 Bluetooth Radio

- Qualified to Bluetooth v5.3 LE specification
- -10dBm ~ 10dBm RF transmit power
- -97dBm (1M) receiver sensitivity
- Digital demodulator for improved sensitivity and co-channel rejection
- Real time digitized RSSI

1.3 Bluetooth Controller

- All device classes support (Broadcaster, Central, Observer, Peripheral)
- All packet types (Advertising / Data / Control)
- Encryption (AES / CCM)
- Bit stream processing (CRC, Whitening)
- Frequency hopping calculation
- Low power modes supporting internal RC oscillator or external 32.768 kHz Crystal oscillator
- Supports power down of the baseband during the protocol's idle periods

1.4 Peripheral Interfaces

- UART port for Debugging and AT Commands
- I2C interface to support external EEPROM or other devices (like G-SENSOR)
- Up to 36 general purpose IOs (34 IOs can be set in interrupt mode)
- General purpose 10-bit SARADC interface, supporting key-press mode and other analog inputs
- 8-channel PWM controller
- General purpose programmable timer for various task
- Watchdog used for tracking unexpected exception

1.5 Integrated Power Control and Regulation

- Embedded Power-On-Reset
- On-chip high efficiency switch-mode, 2.0V to 3.6V input
- On-chip Low Dropout (LDO) Linear Regulator for internal Digital, RF and Analog circuit
- Power management features include software shutdown and hardware wake-up
- Built-in supply voltage detection function

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2. Hardware Information

2.1 Package

2.1.1 FR2011x-U Pinout

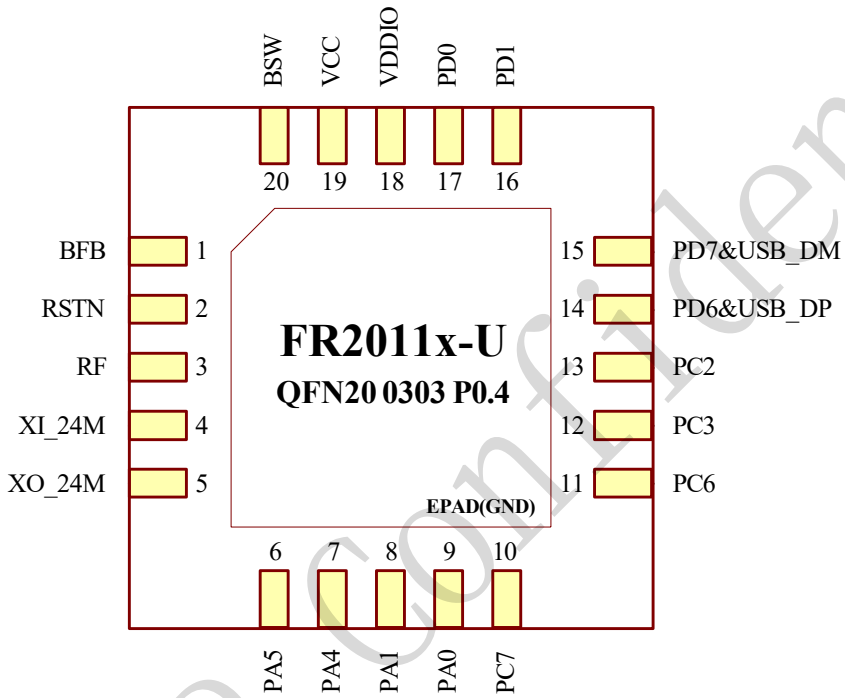


Figure 2-1 FR2011x-U package

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2.1.2 FR2012x-U Pinout

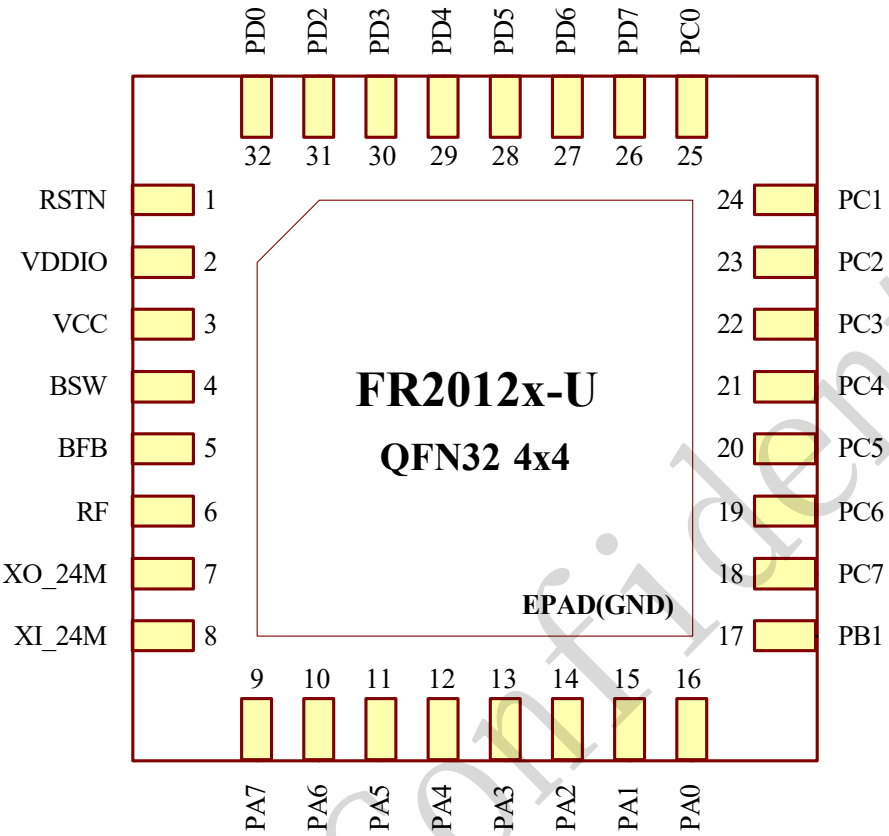


Figure 2-2 FR2012x-U package

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2.1.3 FR2018x-U Pinout

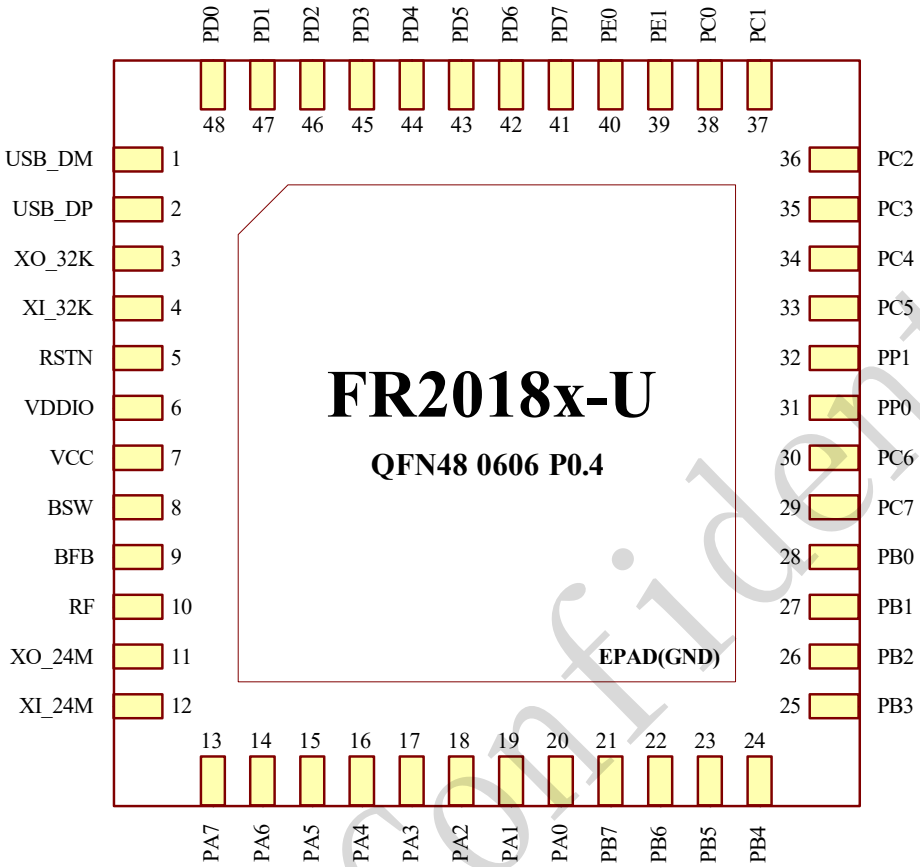
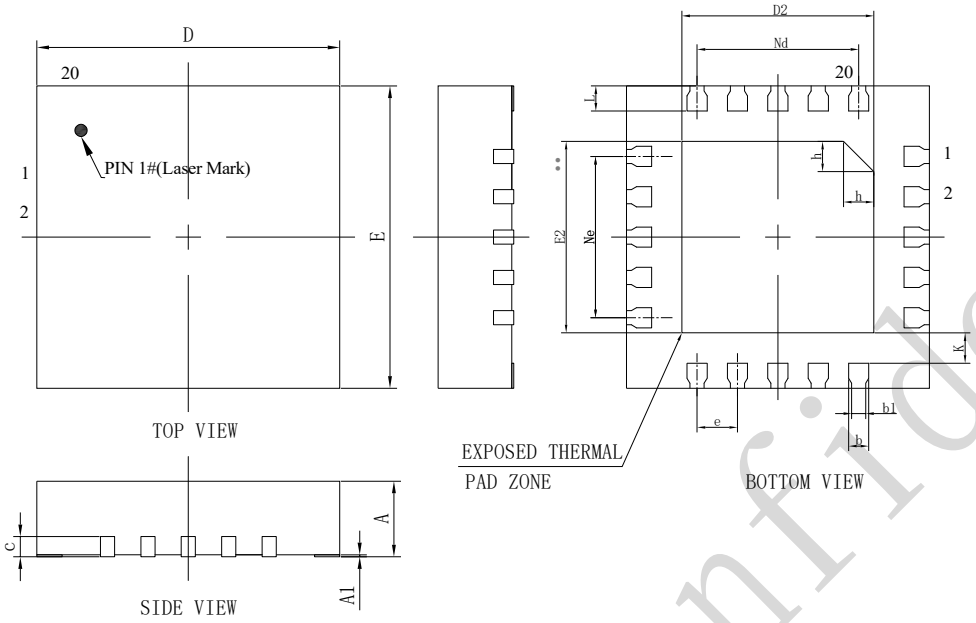


Figure 2-3 FR2018x-U package

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2.2 Package Physical Dimension

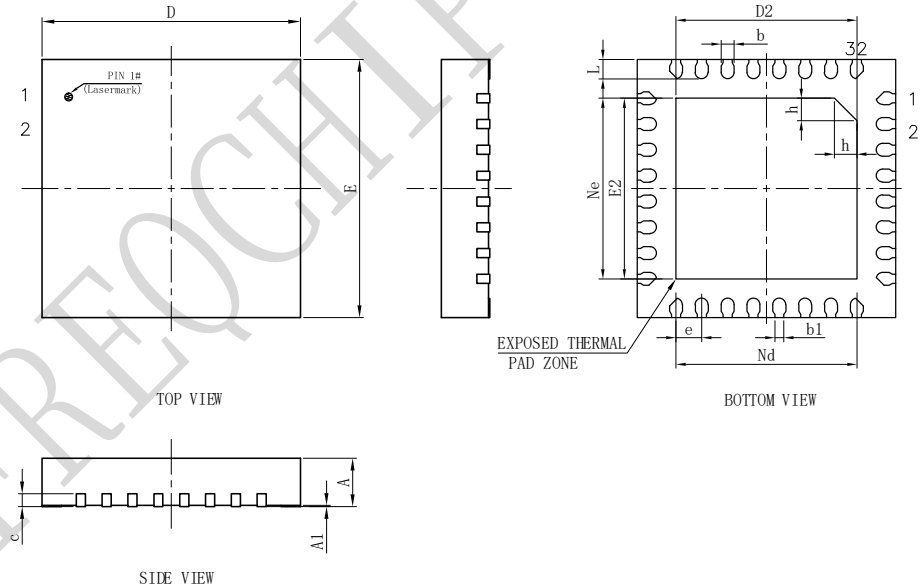
2.2.1 FR2011x-U QFN20



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	0	0.02	0.05
b	0.15	0.20	0.25
b1	0.14REF		
c	0.203REF		
D	2.90	3.00	3.10
D2	1.80	1.90	2.00
e	0.40BSC		
Ne	1.60BSC		
Nd	1.60BSC		
E	2.90	3.00	3.10
E2	1.80	1.90	2.00
L	0.20	0.25	0.30
h	0.25	0.30	0.35
K	0.30REF		

Figure 2-4 FR2011x-U package physical dimensions

2.2.2 FR2012x-U QFN32

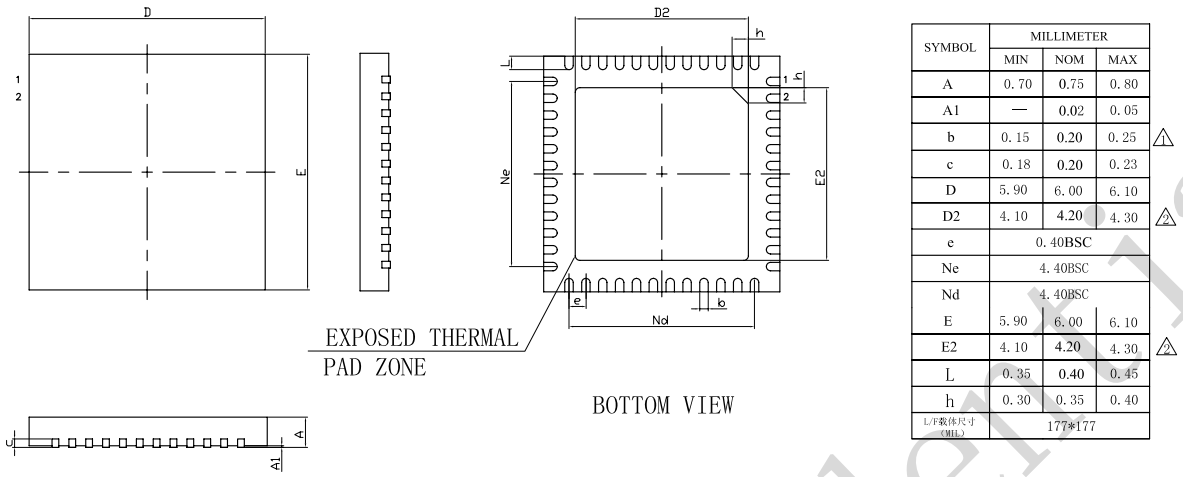


SYMBOL	MILLIMETER			
	MIN	NOM	MAX	
A	0.70	0.75	0.80	
	0.80	0.85	0.90	⚠
	0.85	0.90	0.95	⚠
A1	0	0.02	0.05	
b	0.15	0.20	0.25	
b1	0.14REF			
c	0.18	0.20	0.25	
D	3.90	4.00	4.10	
D2	2.70	2.80	2.90	
e	0.40BSC			
Ne	2.80BSC			
Nd	2.80BSC			
E	3.90	4.00	4.10	
E2	2.70	2.80	2.90	
L	0.25	0.30	0.35	
h	0.30	0.35	0.40	
L/F载体尺寸	122X122			

Figure 2-5 FR2012x-U package physical dimensions

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2.2.3 FR2018x-U QFN48



SYMBOL	MILLIMETER			
	MIN	NOM	MAX	
A	0.70	0.75	0.80	
A1	—	0.02	0.05	
b	0.15	0.20	0.25	1
c	0.18	0.20	0.23	
D	5.90	6.00	6.10	
D2	4.10	4.20	4.30	2
e	0.40BSC			
Ne	4.40BSC			
Nd	4.40BSC			
E	5.90	6.00	6.10	
E2	4.10	4.20	4.30	2
L	0.35	0.40	0.45	
h	0.30	0.35	0.40	
L/F载体尺寸 (MIL)	177*177			

Figure 2-6 FR2018x-U package physical dimensions

2.3 Pins Description

2.3.1 FR2011x-U Pins

table 2-1 FR2011X-U pin description

Pin No.	Pin Name	Type	Description
1	BFB	Analog	BUCK feedback input
2	RSTN	Analog	Chip reset pin, low active
3	RF	Analog	RF input and output. Filter and matching circuits must be reserved between the chip and the antenna.
4	XI_24M	Analog	24MHz Crystal oscillator input
5	XO_24M	Analog	24MHz Crystal oscillator output
6	PA5	I/O	Multi-Function GPIO
7	PA4	I/O	Multi-Function GPIO
8	PA1	I/O	Multi-Function GPIO / UART0_TXD_PROG
9	PA0	I/O	Multi-Function GPIO / UART0_RXD_PROG
10	PC7	I/O	Multi-Function GPIO / SWDIO
11	PC6	I/O	Multi-Function GPIO / SWCLK
12	PC3	I/O	Multi-Function GPIO
13	PC2	I/O	Multi-Function GPIO
14	PD6 & USB_DP	I/O	Multi-Function GPIO / SARADC1 & USB Data Positive

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Pin No.	Pin Name	Type	Description
15	PD7 & USB_DM	I/O	Multi-Function GPIO / SARADC0 & USB Data Minus
16	PD1	I/O	Multi-Function GPIO / SARADC6
17	PD0	I/O	Multi-Function GPIO / SARADC7
18	VDDIO	Analog	IOLDO output, used to supply power to GPIO circuits; place decoupling capacitors in close proximity.
19	VCC	Power	Power supply for chip. Place decoupling capacitors in close proximity.
20	BSW	Analog	BUCK Switch output

2.3.2 FR2012x-U Pins

table 2-2 FR2012x-U pin description

引脚号	引脚名称	类型	描述
1	RSTN	Analog	Chip reset pin, low active
2	VDDIO	Analog	IOLDO output, used to supply power to GPIO circuits; place decoupling capacitors in close proximity.
3	VCC	Power	Power supply for chip. Place decoupling capacitors in close proximity.
4	BSW	Analog	BUCK Switch output
5	BFB	Analog	BUCK feedback input
6	RF	Analog	RF input and output. Filter and matching circuits must be reserved between the chip and the antenna.
7	XO_24M	Analog	24MHz Crystal oscillator output
8	XI_24M	Analog	24MHz Crystal oscillator input
9	PA7	I/O	Multi-Function GPIO
10	PA6	I/O	Multi-Function GPIO
11	PA5	I/O	Multi-Function GPIO
12	PA4	I/O	Multi-Function GPIO
13	PA3	I/O	Multi-Function GPIO
14	PA2	I/O	Multi-Function GPIO
15	PA1	I/O	Multi-Function GPIO / UART0_TXD_PROG
16	PA0	I/O	Multi-Function GPIO / UART0_RXD_PROG
17	PB1	I/O	Multi-Function GPIO
18	PC7	I/O	Multi-Function GPIO / SWDIO

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引脚号	引脚名称	类型	描述
19	PC6	I/O	Multi-Function GPIO / SWCLK
20	PC5	I/O	Multi-Function GPIO
21	PC4	I/O	Multi-Function GPIO
22	PC3	I/O	Multi-Function GPIO
23	PC2	I/O	Multi-Function GPIO
24	PC1	I/O	Multi-Function GPIO / SARADC8
25	PC0	I/O	Multi-Function GPIO / SARADC9
26	PD7	I/O	Multi-Function GPIO / SARADC0
27	PD6	I/O	Multi-Function GPIO / SARADC1
28	PD5	I/O	Multi-Function GPIO / SARADC2
29	PD4	I/O	Multi-Function GPIO / SARADC3
30	PD3	I/O	Multi-Function GPIO / SARADC4
31	PD2	I/O	Multi-Function GPIO / SARADC5
32	PD0	I/O	Multi-Function GPIO / SARADC7

2.3.3 FR2018x-U

table 2-3 FR2018x-U pin description

引脚号	引脚名称	类型	描述
1	USB_DM	I/O	USB Data Minus
2	USB_DP	I/O	USB Data Positive
3	XO_32K	Analog	24MHz Crystal oscillator output
4	XI_32K	Analog	24MHz Crystal oscillator input
5	RSTN	Analog	Chip reset pin, low active
6	VDDIO	Analog	IOLDO output, used to supply power to GPIO circuits; place decoupling capacitors in close proximity.
7	VCC	Power	Power supply for chip. Place decoupling capacitors in close proximity.
8	BSW	Analog	BUCK Switch output
9	BFB	Analog	BUCK feedback input
10	RF	Analog	RF input and output. Filter and matching circuits must be reserved between the chip and the antenna.
11	XO_24M	Analog	24MHz Crystal oscillator output

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引脚号	引脚名称	类型	描述
12	XI_24M	Analog	24MHz Crystal oscillator input
13	PA7	I/O	Multi-Function GPIO
14	PA6	I/O	Multi-Function GPIO
15	PA5	I/O	Multi-Function GPIO
16	PA4	I/O	Multi-Function GPIO
17	PA3	I/O	Multi-Function GPIO
18	PA2	I/O	Multi-Function GPIO
19	PA1	I/O	Multi-Function GPIO / UART0_TXD_PROG
20	PA0	I/O	Multi-Function GPIO / UART0_RXD_PROG
21	PB7	I/O	Multi-Function GPIO
22	PB6	I/O	Multi-Function GPIO
23	PB5	I/O	Multi-Function GPIO
24	PB4	I/O	Multi-Function GPIO
25	PB3	I/O	Multi-Function GPIO
26	PB2	I/O	Multi-Function GPIO
27	PB1	I/O	Multi-Function GPIO
28	PB0	I/O	Multi-Function GPIO
29	PC7	I/O	Multi-Function GPIO / SWDIO
30	PC6	I/O	Multi-Function GPIO / SWCLK
31	PP0	I/O	Multi-Function GPIO
32	PP1	I/O	Multi-Function GPIO
33	PC5	I/O	Multi-Function GPIO
34	PC4	I/O	Multi-Function GPIO
55	PC3	I/O	Multi-Function GPIO
36	PC2	I/O	Multi-Function GPIO
37	PC1	I/O	Multi-Function GPIO / SARADC8
38	PC0	I/O	Multi-Function GPIO / SARADC9
39	PE1	I/O	Multi-Function GPIO
40	PE0	I/O	Multi-Function GPIO
41	PD7	I/O	Multi-Function GPIO / SARADC0
42	PD6	I/O	Multi-Function GPIO / SARADC1
43	PD5	I/O	Multi-Function GPIO / SARADC2
44	PD4	I/O	Multi-Function GPIO / SARADC3
45	PD3	I/O	Multi-Function GPIO / SARADC4
46	PD2	I/O	Multi-Function GPIO / SARADC5
47	PD1	I/O	Multi-Function GPIO / SARADC6

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引脚号	引脚名称	类型	描述
48	PD0	I/O	Multi-Function GPIO / SARADC7

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2.4 Application Circuit

2.4.1 FR2011x-U Circuit

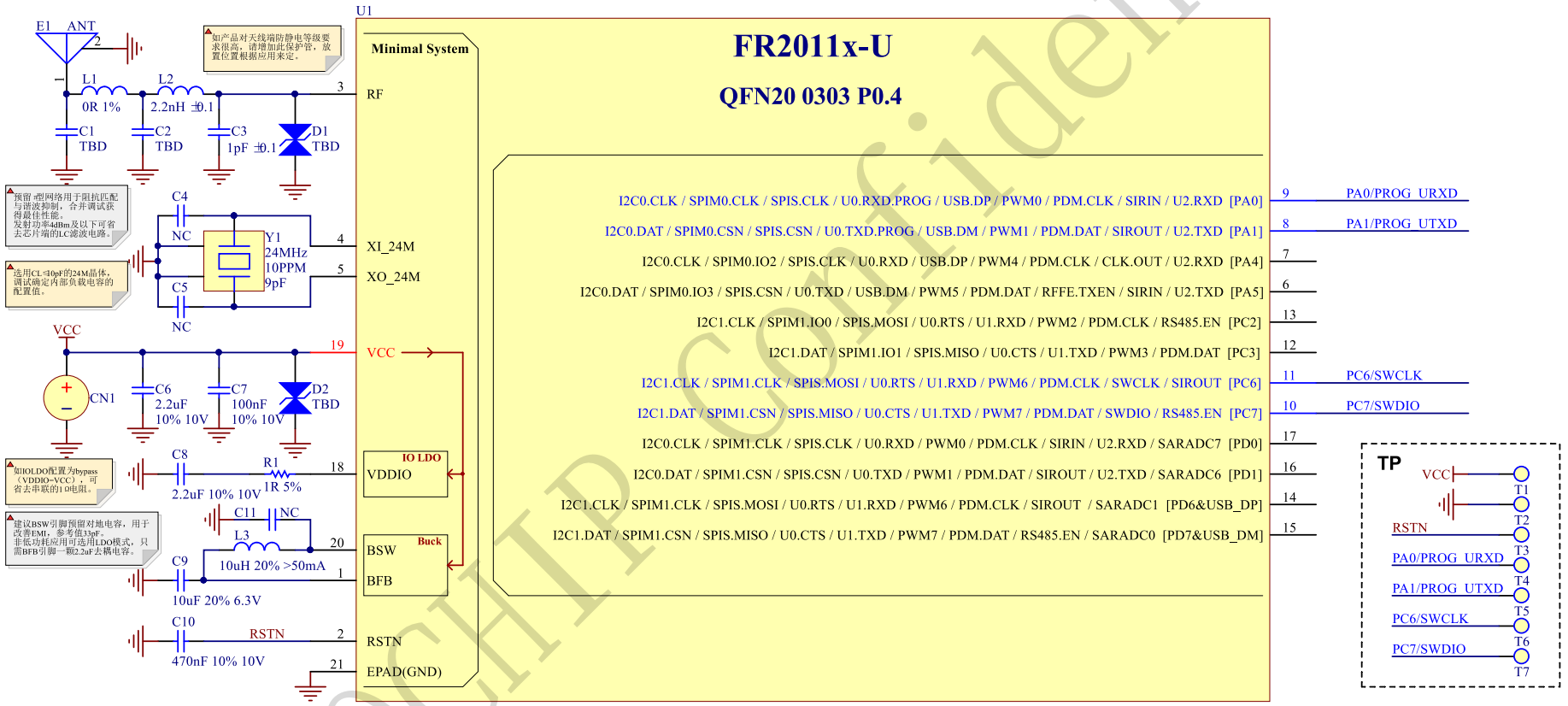


Figure 2-7 FR2011x-U application circuit

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2.4.2 FR2012x-U Circuit

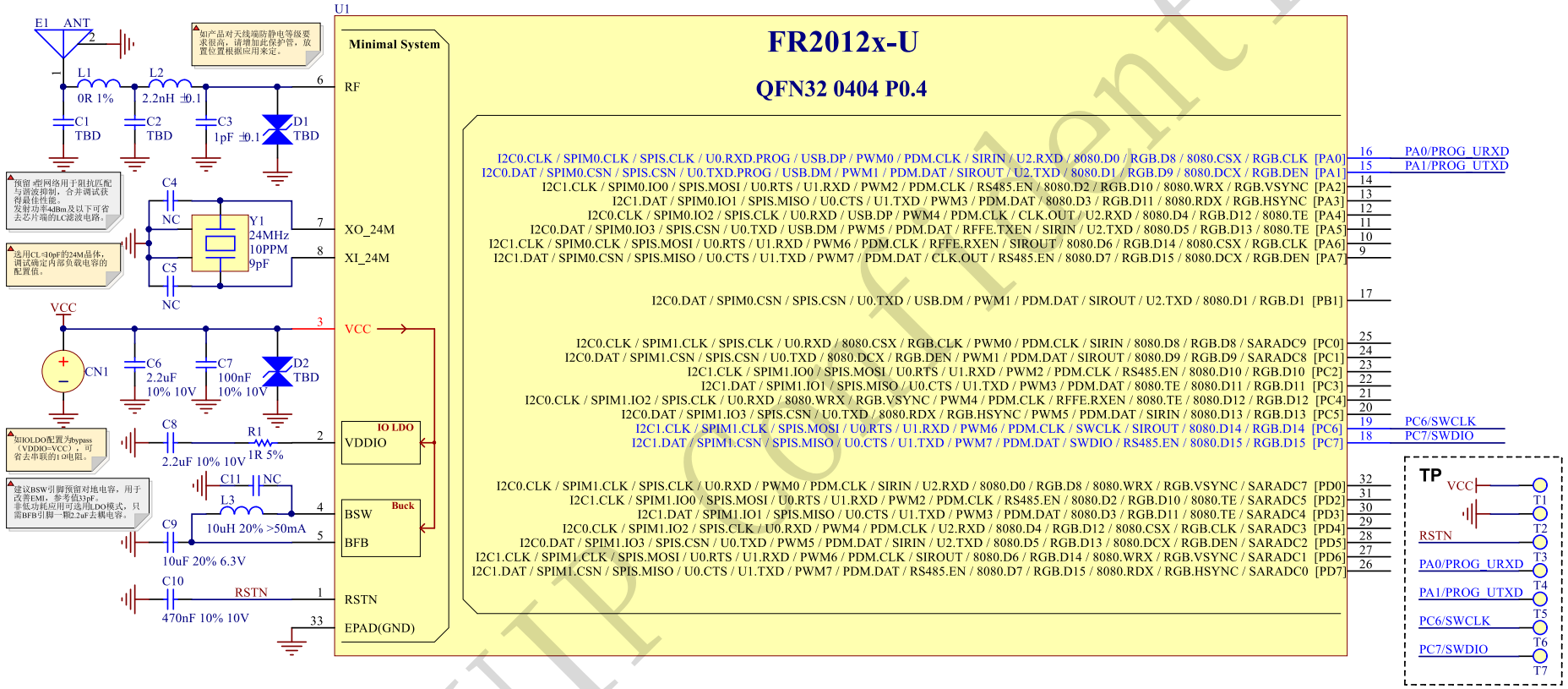


Figure 2-8 FR2012x-U application circuit

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2.4.3 FR2018xP-U Circuit

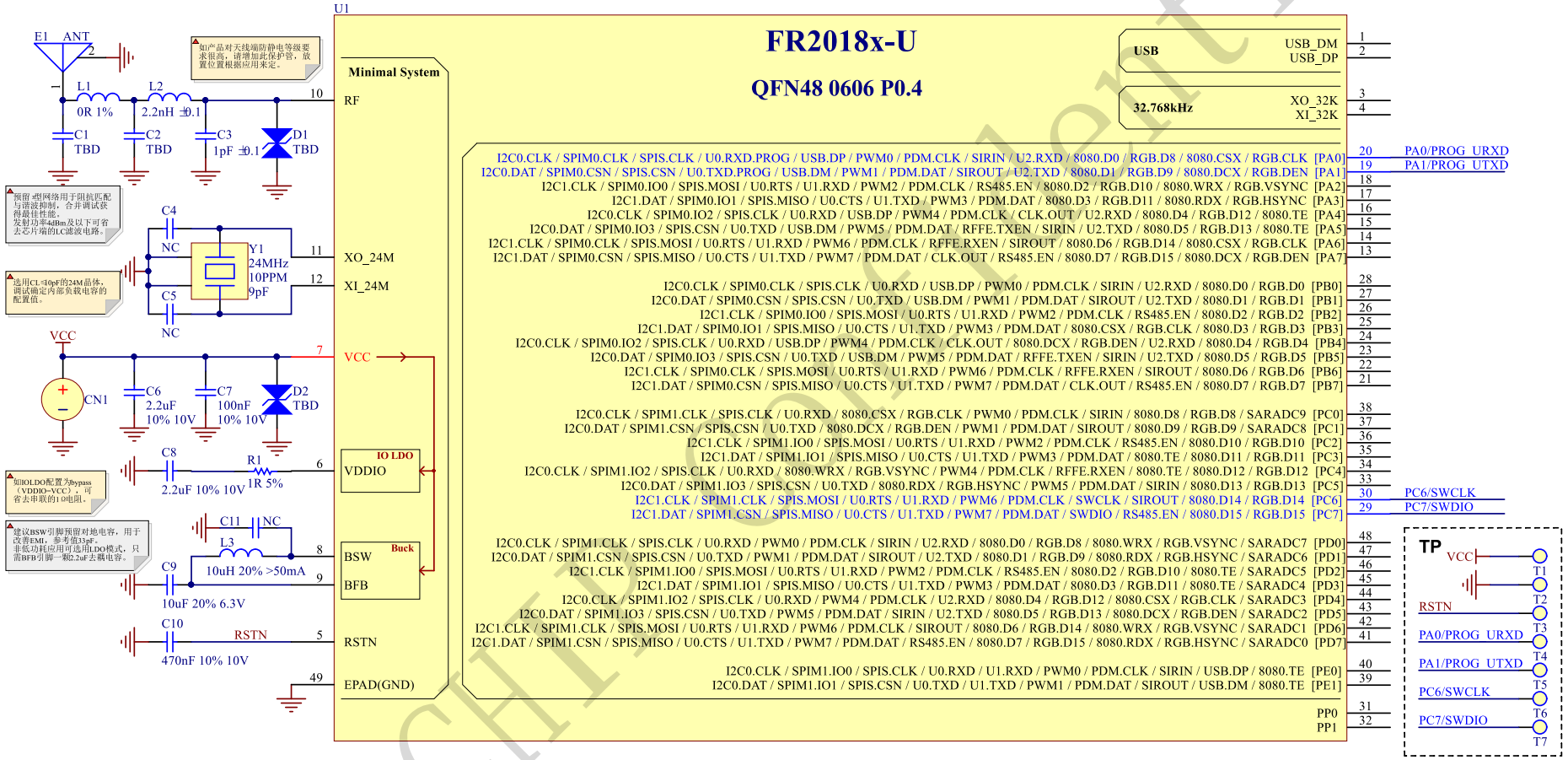


Figure 2-9 FR2018xP-U application circuit

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3. Electrical Characteristics

3.1 Recommended Operating Conditions

Table 3-1 Recommended Operating Conditions

Operating Condition		Min.	Typ.	Max.	Unit
Operating Temperature Range		-40	25	105	℃
I/O Voltage	VDDIO	1.8	3.3	3.6	V
Supply Voltage	VCC	2.0	3.3	3.6	V

3.2 Power Consumption

Table 3-2 Power Consumption

Operation Mode	Average	Max.	Unit
TX peek current (0dB)		7.6	mA
RX peek current		6.01	mA
Sleep mode current	5.4 (include 32K retention RAM)		μA
Power off mode current	2.3		μA
Shutdown mode current	80		nA

3.3 Crystal Oscillator

Table 3-3 Crystal Oscillator

Parameter	Min.	Typ.	Max.	Unit
Clock Frequency	24	24	24	MHz
CL Load capacitance	6	9	10	pF
Tolerance	-10	-	10	PPM
Equivalent Series Resistance (ESR)	-	-	80	Ω
Shunt capacitance	-	-	3	pF

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3.4 IO Resistor

Table 3-4 IO Resistor

Pins	pull-up/down configuration	Resistance Value
PA0~PA7, PB0~PB7 PC0~PC5, PD2~PD7 PE0, PE1	pull-up	8K
	pull-down	6.5K
PC6~PC7, PD0~PD1	pull-up	3K
	pull-down	6.5K
USB_DP, USB_DM	pull-up	1.5K
	pull-down	15K

3.5 ESD

Table 3-5 ESD

Human body model(HBM)	**Charged-device model(CDM)**
±2000V	±500V

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Abbreviations

Abbreviations	Descriptions
ADC	Analog-to-Digital-Converter
GPIO	General Purpose Input Output
PMU	Power Management Unit
OSC	Oscillator

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Contact Information

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Revision History

Feedback:

Freqchip welcomes feedback on this product and this document. If you have comments or suggestions, please send an email to docs@freqchip.com.

Reversion Number	Reversion Date	Description
V0.1.0	2025.11.28	Draft
V0.2.3	2025.12.30	Update pins layout and Application circuit
V0.3.1	2026.02.02	Add FR2011B-U, FR2018A-U
V0.3.2	2026.06.26	Update Application circuit
V0.3.3	2026.07.22	Update Pins Description

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